Objective: Predicting Rain

	(Out look) OL	(Temp) T	(Humidity) H	(Windy) W	(Rain) [Label] R
1	Sunny	Hot	High	weak	No
2	Sunny	mild	high	Strong	No
3	Cloudy	Hot	high	weak	Yes
4	Rainy	mild	high	weak	Yes
5	Rainy	Cool	Normal	weak	Yes
6	Rainy	cool	Normal	Strong	No
7	Cloudy	mild	Normal	Strong	Yes
8	Sunny	Cool	Normal	weak	No

Finding Root Node Using Decision Tree

(1) Entropy of label

Y N

$$= -\left(\frac{4}{8}\log_2\left(\frac{4}{8}\right) + \frac{4}{8}\log_2\left(\frac{4}{8}\right)\right)$$

$$= 1$$

(2) Info gain Of Non Prime Attributes

(i) Outlook

--> Entropy Y N

Sunny: $-\left(\frac{0}{3}\log_2\left(\frac{0}{3}\right) + \frac{3}{3}\log_2\left(\frac{3}{3}\right)\right) = 0.00$

Cloudy: $-\left(\frac{2}{2}\log_2\left(\frac{2}{2}\right) + \frac{0}{2}\log_2\left(\frac{0}{2}\right)\right) = 0.000$

Rainy: $-\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)\right) = 0.918$

CamScanner

(2)

--→ weighted entropy

$= \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.918)$

$= 0.3442$

→ info gain of outlook

$= 1 - 0.3442 = \boxed{0.6558}$

(ii) Temperature

--→ Entropy Y N

Hot $= -\left(\frac{1}{2}\log_2\left(\frac{1}{2}\right) + \frac{1}{2}\log_2\left(\frac{1}{2}\right)\right) = 1$

mild $= -\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)\right) = 0.918$

Cool $= -\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) = 0.918$

→ weighted

$= \frac{2}{8}(1) + \frac{3}{8}(0.918) + \frac{3}{8}(0.918)$

$= 0.938$

→ info gain

$= 1 - 0.938 = \boxed{0.0615}$

(iii) Humidity

--> entropy Y N

High = $-\left(\frac{2}{4}\log_2\left(\frac{2}{4}\right) + \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right) = 1$

Normal = $-\left(\frac{2}{4}\log_2\left(\frac{2}{4}\right) + \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right) = 1$

--> weighted

$= \frac{4}{8}(1) + \frac{4}{8}(1) = 1$

--> info gain

$= 1 - 1 = \boxed{0}$

(iv) Windy

--> entropy Y N

weak = $-\left(\frac{3}{5}\log_2\left(\frac{3}{5}\right) + \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right) = 0.884$

Strong = $-\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) = 0.918$

--> weighted

$\frac{5}{8}(0.884) + \frac{3}{8}(0.918)$

$= 0.896$

--> info gain

$1 - 0.896 = \boxed{0.104}$

(4)

	weighted	info gain
OL	0.3422	0.6558
T	0.938	0.0615
H	1	0
W	0.896	0.109

outlook has greatest info gain so we choose outlook as root node.

Humidity will be cutted.